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NORMAL INTERFACE SURFACE MATING SENSOR

Abstract

A normal interface surface mating sensor includes a power source, a plurality of electrically conductive sensing elements, and at least one of an ammeter or a voltmeter. The voltmeter is connected in parallel with the first terminal and the second terminal of the power source. The plurality of electrically conductive sensing elements is coupled to the first terminal of the power source, is arranged on a first contact face in a pattern such that a spatial location of each sensing element generates an electrical signal when a conductive area on an opposing contact face physically contacts the first contact face at the spatial location of the sensing element contacted. Measurements from the ammeter and the voltmeter together are output as a contact event indicator that includes a conductivity measurement corresponding to the electrical signal generated by the physical contact at a sequence of the NISMSs contacted.

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Background/Summary

TECHNICAL FIELD
[0001] This disclosure relates generally to a sensor for detecting contact between opposing faces that are subject to normal translation. More specifically, this disclosure relates to a normal interface surface mating sensor.
BACKGROUND
[0002] Tracking contact between components due to direct mechanical action or as a result of intrinsic geometric changes is valuable for validating proper component response or for detecting intermittent part interference, etc. Tracking contact events in extreme environments (thermal, fit, pressure or other) precludes the application of a component contact sensor assembly.
SUMMARY
[0003] This disclosure relates to a normal interface surface mating sensor.
[0004] In a first embodiment, an apparatus that includes a power source, a plurality of electrically conductive sensing elements, and at least one of an ammeter or a voltmeter. The power source includes a first terminal and a second terminal. The ammeter is connected to the first terminal of the power source to measure electrical current at the first terminal. The voltmeter is connected in parallel with the first terminal and the second terminal of the power source to measure voltage across the power source. The plurality of electrically conductive sensing elements is coupled to the first terminal of the power source. The plurality of electrically conductive sensing elements is positioned on a first contact face of a surface of an external first body, flush with the first contact face. The plurality of electrically conductive sensing elements is arranged in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS) that generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the external first body at the spatial location of the NISMS. The measurements from the at least one of the ammeter or the voltmeter are output as a contact event indicator that includes a conductivity measurement corresponding to the electrical signal generated by the physical contact at a sequence of the NISMSs with the conductive area on opposing contact face.
[0005] In a second embodiment, a method includes providing a power source including a first terminal and a second terminal. The method includes providing at least one of: an ammeter connected to the first terminal of the power source to measure electrical current at the first terminal; or a voltmeter connected in parallel with the first terminal and the second terminal of the power source to measure voltage across the power source. The method includes providing a plurality of electrically conductive sensing elements coupled to the first terminal of the power source and positioned on a first contact face of a surface of an external first body, flush with the first contact face, and arranged in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS) that generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the external first body at the spatial location of the NISMS. The method includes outputting the measurements from the at least one of the ammeter or the voltmeter as a contact event indicator that includes a conductivity measurement corresponding to the electrical signal generated by the physical contact at a sequence of the NISMSs with the conductive area on opposing contact face.
[0006] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of this disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

[0008] FIG. **1** illustrates an example system that includes a normal interface surface mating sensor (NISMS) and two bodies monitored by the NISMS according to this disclosure;

[0009] FIG. **2**A illustrates a front view of multiple sensing elements that mate with multiple corresponding return elements of an NISMS that are surface mounted on, entrenched in, and embedded within two bodies, respectively, according to this disclosure;

[0010] FIG. **2**B illustrates a portion of FIG. **2**A enlarged to show a close up view of the multiple sensing elements electrically isolated from the first body of FIG. **2**A;

[0011] FIGS. **3**A, **3**B, and **3**C respectively illustrate a side view of the surface mounted, entrenched, and embedded positions of the multiple sensing elements relative to the first body of FIG. **2**A, according to this disclosure;

[0012] FIG. **4**A illustrates an example system that includes an array of NISMSs that detect a contact event with angular misalignment from parallel, according to this disclosure;

[0013] FIG. **4**B illustrates an enlarged view of an array of sensing elements on a contact face of a first body within the system shown in FIG. **4**A, according to this disclosure;

[0014] FIG. **5**A illustrates an example system that includes an asymmetrical pattern of NISMSs that detect a contact event with orthogonal translation-induced offset misalignment from parallel, according to this disclosure;

[0015] FIG. **5**B illustrates the asymmetrical pattern of NISMSs on the two contact faces aligned within the system shown in FIG. **5**A, viewed according to a point of view plane B-B;

[0016] FIG. **6**A illustrates an example system that includes a symmetrical pattern of an NISMS that detects a contact event and determines an orthogonal translation vector as a measurement of offset misalignment from parallel, according to this disclosure;

[0017] FIG. **6**B illustrates the two contact faces aligned within the system shown in FIG. **6**A, sharing the same geometric pattern of sensing elements with different element spacing dimensions, viewed according to a point of view plane C-C;

[0018] FIG. **7** illustrates a front view of a sensing element and an abradable resistive element of an NISMS that detects a contact event and face wear from a conductive second body, according to this disclosure;

[0019] FIG. **8** illustrates a front view of a sensing element and an abradable resistive element of an NISMS that detects a contact event and face wear from a non-conductive second body, according to this disclosure; and

[0020] FIG. **9** illustrates an example method for using an NISMS according to this disclosure.

DETAILED DESCRIPTION

[0021] FIGS. **1** through **9**, described below, and the various embodiments used to describe the principles of the present disclosure are by way of illustration only and should not be construed in any way to limit the scope of this disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any type of suitably arranged device or system.

[0022] Existing contact sensing measurements rely on detecting an applied pressure between contacting surfaces, a distance measurement between contact pairs (e.g., two bodies that contact each other), or a mechanical interaction between contact pairs applied to a discrete contact sensing mechanism (e.g., mechanical switch sensors). The distance measurements between contact pair can be detected using reflectometry techniques and proximity sensors that are inductive, capacitive, or magnetic.

[0023] One of the benefits of a normal interface surface mating sensor (NISMS) is that a measurement ("NISMS measurement") is a direct indication of contact (e.g., a detection event) between contact faces of interest (e.g., being monitored) that are located in the area of the sensing element. The function of the NISMS is similar to a switch based contact sensor. However, the mechanical action that closes the electrical circuit within the NISMS is not caused by a secondary mechanical linkage, rather the direct physical interaction of the contact faces of interest being measured. The size of the contact element within the NISMS can be defined dependent on the lead size, and can even be so small as to be a direct deposition lead on an isolating substrate.

[0024] FIG. **1** illustrates a system **100** that includes a normal interface surface mating sensor (NISMS) **110** and two bodies **120** and **130** monitored by the NISMS **110** according to this disclosure. The embodiment of the system **100** shown in FIG. **1** is for illustration only, and other embodiments could be used without departing from the scope of this disclosure.

[0025] In a particular use case scenario, the first and second bodies **120** and **130** are components of an engine and are confined in a small space that is too small to add contact monitoring equipment or inaccessible for other external measurement. The external surface of the first body **120** includes multiple faces including a first contact face **140**, which can be as a right-side face in an x-z plane according to the coordinate system shown in FIG. **1**. The external surface of the second body **130** includes multiple faces including a second contact face **150**, which can be as a left-side face in an x-z plane according to the coordinate system shown. The two contact faces **140** and **150** can be spaced apart by a gap and face each other. The two contact faces **140** and **150** are subject to normal translation, for example along the y-axis, as the arrows of movement **122** and **132** show. In other words, a vector of relative motion between the two bodies **120** and **130** is normal (e.g., orthogonal) to the two contact faces **140** and **150**. Movement **122** of the first body **120** can expand or shrink the gap that defines a separation distance between the two contact faces **140** and **150**. Analogously, movement **132** of the second body **130** can expand or shrink the gap. In this engine scenario, the first and second bodies **120** and **130** are structures are not expected to maintain constant physical contact with each other during normal operation, but can in fact physically contact each other. An owner of the engine or personnel of an operations and maintenance team might desire to know when the first and second bodies **120** and **130** physically contact, or more specifically, when the first and second contact faces **140** and **150** physically contact each other. The two contact faces **140** and **150** can have dimensions that are substantially the same, such that when the two contact faces **240** and **250** are parallel aligned without (angular or offset) misalignment, the entire area of the first contact face **140** is flush in physical contact with the entire area of the second contact face **150**. Note that while an engine-based scenario is specifically described as one type of example use case of the NISMS **110**, the NISMS **110** can be used in any suitable applications (e.g., use cases) in any suitable environments.

[0026] The first body **120** can be electrically conductive or non-conductive, as embodiments of this disclosure are independent from the conductivity of material of the first body **120**. In the system **100**, and the second body **130** is electrically conductive, for example, composed of metal or other electrically conductive material. However, the second body can be non-conductive in a different system (other than system **100**) in a different embodiment of this disclosure, as shown in FIG. **8**.

[0027] The system **100** is an embodiment in which the NISMS **110** is a device to be installed into a position relative to an external device (e.g., engine), and the external device includes the first body **120** and second body **130**. That is, the NISMS **110** can be installed into a position relative to the two bodies **120** and **130** that enables the NISMS **110** to detect a contact event that includes two contact faces **140** and **150** in physical contact. Also, the position where the NISMS **110** is installed enables (e.g., does not interfere with or influence) movements **122** and **132** of two bodies **120** and **130**, respectively. In a different system (other than system **100**) in a different embodiment of this disclosure, the NISMS **110** can be integrated into with the first body, with the second body, or with both bodies of a device (e.g., engine) that the NISMS **110** monitors, as shown in FIGS. **3**B and **3**C.

[0028] The NISMS **110** is composed of an electrically isolated (i.e., electrically isolated from the first body **120**) sensing element **160** embedded in or otherwise attached to a body (for example, the first body **120**) such that the sensing element **160** is flush, recessed or proud of a proximal face (e.g., first contact face **140**) of the body anticipated to make contact with a mating face (e.g., second contact face **150**). An electric insulator **262**, which is composed from one or more electric insulator materials that have high resistivity and prevent the flow of electric current, is disposed between the sensing element **160** and the surface of the first body **120**, and can surround the surface of the sensing element **160** except for areas of the electric insulator **262** that face the second contact face **150** of the second body **130**. The sensing element **160** is electrically conductive, such as a conductive probe. The sensing element **160** is a single electrical node in the system **100**, but can include multiple electrical nodes in a different embodiment of this disclosure, as shown in FIGS. **2** and **4-6**.

[0029] The NISMS **110** includes a power source (battery) **170** that includes a first terminal **172** and a second terminal **174**. The NISMS **110** additionally includes an ammeter **180** and/or a voltmeter **190**. The battery **170** and the voltmeter **190** are connected in an electric parallel circuit, and the voltmeter **190** measures voltage (i.e., electrical potential) between the first and second terminals **172** and **174**. The first terminal **172** of the battery **170** is electrically connected to the sensing element **160**, and the ammeter **180** measures electrical current at the first terminal **172**. In the system **100**, the second terminal **174** of the battery **170** is electrically connected to the second body **130**. For example, the second terminal **174** can be physically and electrically coupled to the top face **152** of the second body **130**.

[0030] The NISMS **110** detects when the first contact face **140** and the second contact face **150** are in physical contact, which physical contact is primarily due to the movement **122**, **132** of opposing body faces in a vector normal to the respective faces (i.e., a closing of the gap between the contact faces **140** and **150**). In the system **100**, the sensing element **160** is surface mounted at a position on an outer surface of the first body **120**, for example, a top face **142** of the first body **120** in an x-y plane according to the coordinate system shown in FIG. **1**. The position of the sensing element **160** is flush with (the plane of) the contact face **140**. That is, the sensing element **160** can be surface mounted onto any outer surface of the first body **120**, so long as the sensing element **160** switches into closed-circuit electrical contact with the second terminal **174** of the battery **170** upon physical contact between the two contact faces **140** and **150**. Various embodiments of this disclosure change the position of the sensing element **160** relative to the first body **120** or change the shape of the sensing element **160**, however, the positioning of the sensing element **160** remains such that the contact of a proximal face to a distal face will result in an electrically conductive contact being made between sensing element **160** and the distal face.

[0031] The second contact face **150** (e.g., distal face), upon contacting the sensing element **160**, forms a completed circuit through electrical conduction through the second body **130** (i.e., distal body) of the system **100**. In a different embodiment of this disclosure, the distal face, upon contacting the sensing element **160**, forms a completed circuit through a provisioned return path applied to the distal face and distal body, for example, as shown in FIG. **2**A.

[0032] The NISMS **110** detects a contact event based on a change in resistance in the sensing circuit, such as through a resistance change in the circuit. Continuity between the sensing element **160** and the second body **130** changes the sensor measurements indicated as potential (V) of the circuit, as load (A) of the circuit, or as both voltage and current (V and A) of the circuit. That is, embodiments of this disclosure do not require a combined use of the ammeter and the voltmeter, as either is sufficient to indicate an anticipated change in the circuit caused upon contact at the sensing element **160**. It is understood that some embodiments of this disclosure, both the ammeter and the voltage meter used together (e.g., concurrently) to generate the sensor measurements changed, which indicate change in the circuit caused upon contact at the sensing element **160**. Another way that the NISMS **110** detects the contact event is based on a change in an externally applied potential as the circuit alternates between electrically open state and electrically open closed states. A current measurement output **182** from the ammeter **180** with a voltage measurement output **192** from the voltmeter **190** can be used to measure the change in resistance (e.g., a conductivity change) in the sensing circuit, for example, the resistance change can be relative to a reference resistance value. Particularly, the ammeter **180** and voltmeter **190** can measure a direct or an alternating electric current.

[0033] FIG. **2**A illustrates a front view of multiple sensing elements **260***a*-**260***c* that mate with multiple corresponding return elements **270***a*-**270***c* of an NISMS that are surface mounted on, entrenched in, and embedded within two bodies, respectively, according to this disclosure. As shown in FIG. **2**A, a system **200** includes a first body **220** and a non-conductive second body **230**. The first body **220** can be the same as or similar to the first body **120** of FIG. **1**. The second body **230** is composed of a non-conductive material that is different from the conductive material of which the second body **130** of FIG. **1** is composed, but otherwise can be similar to the second body **130**.

[0034] FIG. **2**B illustrates a portion of FIG. **2**A enlarged to show a close up view of the multiple sensing elements **260**A-**260**C electrically isolated from the first body **220** by an electric insulator **262** of FIG. **2**A. The electric insulator **262** can be similar to and can perform the same function as the electric insulator of **162** of FIG. **1**.

[0035] In the system **200**, a first pair of mating elements (labeled A) includes a first sensing element **260**A that is adapted to mate (i.e., mates with) a corresponding first return element **270**A. A second pair of mating elements (labeled B) and a third pair of mating elements (labeled C) include a second and a third sensing element **260**B and **260**C that are adapted to mate with a corresponding second and third return element **270**B and **270**C, respectively. When the two contact faces **240** and **250** of the two bodies are parallel aligned without (angular or offset) misalignment, each of the pairs of mating elements A, B, and C is mated such that a face of the sensing element is in physical contact with an opposing face of the corresponding return element, for example, with opposing faces flush to each other. In some embodiments, a pair of mating elements (A, B, or C) have a female-male shapes such that one among the pair of mating elements has a recessed female shape and the other among the pair of mating elements has a protruded male shape corresponding to the recessed female shape. The female and male shapes of the pair of mating elements physically contact and are mated when the two contact faces **240** and **250** are parallel aligned without (angular or offset) misalignment.

[0036] FIGS. **3**A, **3**B, and **3**C respectively illustrate a side view of the surface mounted, entrenched, and embedded positions of the multiple sensing elements relative to the first body **220** of FIG. **2**A, according to this disclosure. The side view is from the point of view of the cutting plane line A-A shown in FIG. **2**A.

[0037] The first sensing element **260**A is surface mounted on the outer surface of the first body **220**, for example, on a top face **242** of the first body **220** as shown in FIG. **3**A. With a mirror-image similarity, the first return element **270**A is surface mounted on the outer surface of the second body **230**, for example, on a top face **252** of the second body **230**.

[0038] The second sensing element **260**B is entrenched in the first body **220**, as shown in FIG. **3**B. For example, the outer surface of the first body **220** can include a groove, and the second sensing element **260**B can be installed into the groove to fill the groove. After installed within the groove, a portion of the second sensing element **260**B is faces the groove and is positioned internal to the groove, another portion **264** of the second sensing element **260**B is exposed outside of the groove and protrudes (e.g., in a direction along the z-axis) beyond the surface of the first body **220** to be exposed outside of the first body **220**. For example, the second sensing element **260**B can be a lead (such as a lead on a printed circuit board) formed integral with the first body **220** and into the groove, or can be formed separate from the first body **220** and later inserted into the groove of the first body **220**.

[0039] The third sensing element **260**C is embedded within the first body **220**, As shown in FIG.

3C. That is, third sensing element **260**C is formed integrally with the first body **220**, and is positioned internal to the first body **220**. An inner surface of the electric insulator **262** surrounds the third sensing element **260**C. An outer surface of the electric insulator **262** is surrounded by and is positioned internal to the first body **220**, to electrically isolate the third sensing element **260**C from the first body **220**.

[0040] FIG. **4**A illustrates an example system **400** that includes an array **402** of NISMSs that detect a contact event with angular misalignment from parallel, according to this disclosure. FIG. **4**B illustrates an enlarged view of an array of sensing elements on a contact face of a first body within the system shown in FIG. **4**A, according to this disclosure. FIGS. **4**A and **4**B are described together as FIG. **4**. The array **402** of multiple sensing elements **460_1** through **460_4** is a component of an NISMS that detects a contact event with angular misalignment from parallel, according to this disclosure. As shown in FIG. **4**, a system **400** includes a first body **420** and a non-conductive second body **430**. The first body **420** and the second body **430** can be the same as or similar to corresponding components **120** and **130** of FIG. **1**. The second body **430** is composed of an electrically conductive material. The two contact faces **440** and **450** are subject to normal translation, for example, along the y-axis, as the arrows of movement **422** and **432** show.

[0041] The multiple NISMS sensing elements **460_1** through **460_4** (generally, **460**) are applied (e.g., in the pattern of the array) as the array **402** to one contact face (**440** or **450**) among a pair of contact faces **440** and **450**. Two sensing elements **460** can be used to indicate a closure angle θ (e.g., angle of angular misalignment) between the opposing two contact faces **440** and **450**.

[0042] The NISMS detects a contact event upon contact made between the two faces contact faces **440** and **450** at a single sensing element **460** or at multiple sensing elements **460** of the array **402** concurrently. That is, the arrayed sensing element **460** will conduct electricity in a closed circuit, indicting local contact event detected. The ammeter **180** and/or voltmeter **190** can measure the electricity conducted in the closed circuit, enabling the NISMS to detect the contact event.

[0043] The spatial location of the sensing elements **460** within the array **402** together with a sequence of the conductivity change within the NISMS causes the NISMS to qualitatively indicate an angular deviation from parallel between the contact face pair **440** and **450**. For example, the conductivity change can be measured by and output from the ammeter and voltmeter of FIG. **1**.

[0044] For example, the NISMS can qualitatively indicate an angular misalignment from parallel in a particular direction of rotation. For example, when the third and fourth sensing elements **460_3** and **460_4** conduct electricity in a closed circuit, the NISMS can indicate a rotation of the second contact face **450** about a first hinge line H**1**, which is formed based on the spatial location of the third and fourth sensing elements **460_3** and **460_4**. In this example, the contact events detected concurrently at the third and fourth sensing elements **460_3** and **460_4** (while the remaining sensing elements **460_1** and **460_2** remain open circuited without detection of a contact event) cause the NISMS to output an indicator that the second face **450** is rotated about H**1** and offset in the direction of the x-z plane. The table below shows that the NISMS can indicate angular misalignment in the particular direction of rotation about other hinge lines H**2**, H**3**, and H**4**, as shown in Table 1.

TABLE-US-00001 TABLE 1 Plane of Angular Axis of Contact Event Currently Detected? Misalignment Rotation 460_1 460_2 460_3 460_4 y-z H1 NO NO YES YES x-y H2 YES NO YES NO y-z H3 NO YES NO YES y-z H4 YES YES NO NO

[0045] The NISMS can include angular offset between the two contact faces **440** and **450** produces asynchronous continuity for sensing elements **460_1** through **460_4** as the faces make contact. By mapping continuous circuits for sensing elements, as shown in Table 1, a range of motion (ROM) relative angle of the pair of contact faces **440** and **450** can be determined based on which sequence of sensing elements **460_1** through **460_4** are conducting an electric signal.

[0046] FIG. **5**A illustrates an example system **500** that includes an asymmetrical pattern **502** of NISMSs **510***a*-**510***g* that detect a contact event with orthogonal translation-induced offset misalignment from parallel, according to this disclosure. FIG. **5**B illustrates the asymmetrical pattern of NISMSs on the two contact faces aligned within the system shown in FIG. **5**A, viewed according to a point of view plane B-B. FIGS. **5**A and **5**B are described together as FIG. **5**. The system **500** includes at least one continuity NISMS **510**A that includes a continuity sensing element **560***a* positioned on the first contact face **540** at a spatial location that is an origin point **512**. The continuity NISMS **510**A includes a continuity return element **570***a* positioned on the second contact face **550** at a spatial location that corresponds to (e.g., is centered about, and that overlaps) the origin point **512**. When the two contact faces **540** and **550** physically contact in parallel alignment (without angular misalignment and without offset misalignment), the spatial locations of the continuity sensing element **560***a* and continuity return element **570***a* overlap at the origin **512** and electrically connect with each other as an aligned pair of elements. In a scenario where one or both contact faces (e.g., **540** and **550**) are non-conducting, one or more aligned pair of elements can be applied to enable such continuity NISMS **510***a* to generate an electric signal as an indicator of a contact event at the origin **512**.

[0047] To detect orthogonal translation-induced offset misalignment from parallel, the system **500** includes a plurality of NISMSs **510**A-**510**G arranged such that fewer than all pairs of elements are able to create a conducting circuit upon face contact. For example, the NISMSs **510**A-**510**G within the system **500** include a set of multiple sensing elements **560***a*-**560***g* of an NISMS arranged in a first pattern **504** on the first contact face **540** of the first body **520**, and a set of multiple corresponding return elements **570***a*-**570***g* arranged on the second contact face **550** of the second body **530** in a second pattern **506** different from the first pattern **504**, according to this disclosure. If the two parallel faces **540** and **550** shift orthogonally (e.g., in a direction shown by 2D axes **514**) relative to the body translation vector **522** and **532**, the sequence (e.g., set) of sensing elements that are conducting an electric signal will change, indicating a translation in the face plane. If the parallel faces **540** and **550** are aligned, as shown in FIG. **5**, then the sensing elements **560***b*-**560***g* and corresponding return elements **570***b*-**570***g* that form the non-aligned pairs, respectively, will not overlap and will form an open switch that is not conducting an electric signal.

[0048] For example, if second contact face **550** shifts orthogonally (e.g., a positive z direction) such that the non-aligned pair of elements **560***b* and **570***b* overlaps at the spatial location of the second sensing element **560***b*, then the second NISMS **510***b* outputs an electrical signal that is an indicator of orthogonal translation-induced offset misalignment from parallel in the positive z direction. That is, the non-aligned pair of elements **560***b* and **570***b* electrically contact each other to conduct an electric signal between the first and second terminals of a power source. The orthogonal shift in the face plane does not disconnect the continuity NISMS **510***a*, which continues conducting an electrical signal that indicates parallel alignment.

[0049] Asymmetrical arrangements of sensing elements cause electric signal generation based on relative translation of faces **540** and **550**. Resolution of the angle of misalignment and extent of face translation can be changed based on number and relative location of sensing element pairs.

[0050] FIG. **6**A illustrates an example system **600** that includes a symmetrical pattern of an NISMS that detects a contact event and determines an orthogonal translation vector as a measurement of offset misalignment from parallel, according to this disclosure. FIG. **6**B illustrates the two contact faces aligned within the system shown in FIG. **6**A, sharing the same geometric pattern of sensing elements with different element spacing dimensions, viewed according to a point of view plane C-C. FIGS. **6**A and **6**B are described together as FIG. **6**. For example, the plurality of NISMSs **610***a*-**610***e* includes a geometric pattern **602** of multiple sensing elements **660***a*-**660***e* that mate with multiple corresponding return elements **670***a*-**670***e* arranged in the same geometric pattern with a different element spacing dimension. For example, the spacing dimension of the sensing elements **660***a*-**660***e* can be a first distance d**1** between the sensing elements, and the spacing dimension of the return elements **670***a*-**670***e* can be a second distance d**2** between the return elements that is greater than the first distance d**1**.

[0051] For example, the geometric pattern **602** includes an array, which includes a first row, a second row, a first column, a second column, which can be similar to the array pattern in the system **400** of FIG. **4**, and can be rotated (e.g., 45°). The geometric pattern **602** additionally includes a center point where an origin **612** of a continuity NISMS **610***a* (including an aligned pair of elements **660***a* and **670***b*) is located, for example, the geometric pattern **602** can resemble 5-points on a rolling-die, or can be referred to as a quincunx pattern. The diameters of the continuity pair of elements **660***a* and **670***b* can be different from each other, for example, the diameter of the first return element **670***a* can be greater than the diameter of the first sensing element **660***a*. The diameters of the arrayed sensing elements **660***b*-**660***e* (other than the continuity pair at the origin **613**), which form the rows and columns of the array of sensing elements, can be equivalent to each other. The diameters of the arrayed return elements **670***b*-**670***e*, which form the rows and columns of the array of return elements, can be equivalent to each other, and can be the same as or different from the diameter of the arrayed sensing elements.

[0052] To determine an orthogonal translation vector, a specified partial overlap area of a cross sectional area of the arrayed sensing element **660***b*-**660***e* partially overlaps and is in physical and electrical contact with a corresponding arrayed return element **670***b*-**670***e* when the two contact faces **640** and **650** are parallel aligned and in physical contact. A reference resistance measurement or reference conductivity measurement can be defined based on the electrical signals generated by the set of arrayed NISMSs **610***b*-**610***e* partially overlapping, in addition to the electrical signal from the overlap within the continuity NISMS **610***a* at the origin **613**. A translation vector can be determined based on the measured resistance at each NISMS element. The change in resistance measurement (R) is resultant from the changing cross-sectional area of the NISMS elements. For example, the change in resistance measurement (R) is relative to the reference resistance measurement and can be estimated based on Equation 1 below, where R denotes resistance, L denotes length of the material measured, and A denotes a cross-sectional area of the material overlapped.

$$R \approx L/A \qquad (1)$$

[0053] In some embodiments, the continuity NISMS **610***a* maintains a constant (e.g., invariant to translation) contact area, which invariance can be used as a basis upon which to calibration or correct temperature induced resistance change.

[0054] FIG. **7** illustrates a front view of a system **700** that includes a sensing element **760** and an abrasion sensing element **780** (also referred to as an abradable resistive element) of an NISMS that detects a contact event and face wear from a conductive second body, according to this disclosure. The abrasion sensing element **780** is composed of an abradable resistive material that reduces cross-sectional length (L) in response to abrasion from physical contact with the second body **730**. The abradable resistive material can be carbon-ceramic composite, amorphous carbon, or other material. The second body **730** is composed of conductive material and can be the same as or similar to the second body **130** of FIG. **1**. In some embodiments, the second body **730** can be composed of non-conductive material, in which case, a conducting return path can be provided for the second body similar to the return paths associated with the return elements **270***a*-**270***c* of FIG. **2**A.

[0055] When the second contact face **750** physically contacts the abrasion sensing element **780** at the first contact face **740**, both contact and face wear (e.g., abrasion) can be determined. A length (L) of the abrasion sensing element **780** shortens due to abrasion from a previous length L**1** to a current length L**2**. The change of length (ΔL) of the abrasion sensing element **780** represents a change of length of the first body **720** before face wear to a current length of the shortened first body **720′**. The face wear is determined based on a change in resistance of the sensing element **760** that includes the abrasion sensing element **780**. The change in resistance (ΔR) of the abrasion sensing element **780** also represents a change in resistance at the spatial location of the sensing element **760**. The resistance measurement corresponding to the spatial location of the sensing element **760** can be estimated according to Equation 2, where the change of length (ΔL) of the abrasion sensing element **780** is determined based on L**2** subtracted from L**2**, and where the cross sectional area of the abrasion sensing element **780** is denoted as A. In various embodiments of this disclosure, the cross sectional area of the abrasion sensing element **780** can be equivalent to or greater than the cross sectional area of a lead (composed of lead material) that forms part of the sensing element **760**.

$$\Delta R \approx \Delta L / A \qquad (2)$$

[0056] FIG. **8** illustrates a front view of a sensing element **860** and an abradable resistive element **780** of an NISMS that detects a contact event and face wear from a second body **830**, according to this disclosure. That is, two contact faces **840** and **850** of the first body **830** and the second body **830** physically contact when the bodies exhibit movement in a direction normal to a face plane. The second body **830** is composed of conductive material and can be the same as or similar to the second body **130** of FIG. **1**. For example, the second body **830** is composed of a conductive material, but is unable to support installation of return element onto the second contact face **850**. In some embodiments, the second body **830** is unable to support a surface mounted return path installed onto another face of the second body **830**, other than the second contact face **850**. Installation of the NISMS **810** includes installing (via surface mounting, entrenching, embedding) a return element **870** (onto the first body **820** that also includes the sensing element **780** of FIG. **7** installed. The cross sectional area of the abrasion sensing element **780** can be greater than the cross sectional area of the sensing element **860** that is a lead, and can be greater than the cross sectional area of a second lead **870** that forms a return path **870**. In some embodiments, the abradable resistive element **780**, which is electrically coupled to both the return path **870** and to the path of the sensing element **860** lead, performs a similar function to both functions of a pair of aligned elements (e.g., **560***a* and **570***a* of FIG. **5**). The shortened first body **820′** can be shortened similar to the shortened first body **720′** of FIG. **7**.

[0057] FIG. **9** illustrates an example method **900** for using a normal interface surface mating sensor according to this disclosure. The method **900** can be performed by a computer processor of automated manufacturing system that is configured to install an NISMS onto or embed an NISMS into a first body and/or the second body of FIGS. **1-8**. The processor can use or control other components of the automated manufacturing system to perform the method **900**, however, for simplicity, the method **900** is described as performed by the processor.

[0058] At block **902**, the processor provides a power source including a first terminal and a second terminal. At block **904**, the processor provides an ammeter connected to the first terminal of the power source to measure electrical current at the first terminal. At block **904**, the processor provides a voltmeter connected in parallel with the first terminal and the second terminal of the power source to measure voltage across the power source.

[0059] At block **906**, the processor provides a plurality of electrically conductive sensing elements coupled to the first terminal of the power source and positioned on a first contact face of a surface of an external first body, flush with the first contact face. The plurality of electrically conductive sensing elements is arranged in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS) that generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the first body at the spatial location of the NISMS.

[0060] As part of providing the plurality of sensing elements, at block **908**, the processor arranges the plurality of sensing elements in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS). Each NISMS generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the first body at the spatial location of the NISMS.

[0061] At block **912**, measurements from the ammeter and the voltmeter together are conveyed to an input interface of a processor that performs a function based on the received measurements. Particularly, the NISMSs that are conducting an electric signal (referred to as a sequence of NISMISs) output measurements from the ammeter and the voltmeter together as a contact event indicator. The input interface of the processor receives the measurements output from the sequence of NISMSs as the contact event indicator, which includes a conductivity measurement corresponding to the electrical signal generated by the physical contact at a sequence of the NISMSs with the conductive area on opposing contact face.

[0062] Although FIG. **9** illustrates an example method **900** for using an NISMS, various changes may be made to FIG. **9**. For example, while shown as a series of steps, various steps in FIG. **9** could overlap, occur in parallel, occur in a different order, or occur any number of times.

[0063] As a particular example, in some embodiments, the pattern includes an array including a first row, a second row, a first column, and a second column. The conductivity measurement corresponding to first and second NISMSs as the sequence of the NISMSs in the physical contact with the second body indicates angular misalignment about a first axis of rotation. The conductivity measurement corresponding to first and third NISMSs as the sequence of the NISMSs in the physical contact with the second body indicates angular misalignment about a second axis of rotation orthogonal to the first axis. The conductivity measurement corresponding to third and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second body indicates angular misalignment about a third axis parallel to the first axis. The conductivity measurement corresponding to second and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second body indicates angular misalignment about a fourth axis parallel to the third axis.

[0064] In some embodiments, the method **900** further comprises providing a set of return elements asymmetrically arranged in the pattern and positioned on the opposing contact face of the second body. The set of asymmetrically arranged return elements includes a continuity return element characterized by a wider diameter than non-aligned return elements among the set of return elements. The plurality of sensing elements is asymmetrically arranged in the pattern. When the opposing contact face physically contacts the first contact face, the continuity return element maintains electrical contact with a first continuity sensing element among the plurality of sensing elements while any non-aligned return element electrically contacts a corresponding non-aligned sensing element as a non-aligned pair. The contact event indicator output includes a first indicator of translation in a first direction along a first axis when a first non-aligned pair are contacting. The contact event indicator output includes a second indicator of translation in a second direction along the first axis when a second non-aligned pair are contacting, the first and second directions opposite to each other. The contact event indicator output includes a third indicator of translation in a third direction along a second axis orthogonal to the first axis when a third non-aligned pair are contacting. The contact event indicator output includes a fourth indicator of translation in a fourth direction along a fourth axis orthogonal to the second axis when a fourth non-aligned pair are contacting, the third and fourth directions opposite to each other. The embodiment could be further include the processor determining, based on the contact event indicator output, an angle of translation and distance of translation from an origin where the spatial locations of the continuity sensing element and the continuity return element overlap, based on: at least one of the first, second, third, or fourth indicators; and the origin relative the spatial locations of the sequence of the NISMSs among the non-aligned pairs contacting.

[0065] In some embodiments, the processor is further configured to provide an abrasion sensing element attached to at least one sensing element among the plurality of sensing elements. The abrasion sensing element is composed of an abradable resistive material that reduces cross-

sectional length in response to abrasion from physical contact with the second body. The method can further include outputting (or conveying to the input interface of the processor) an abrasion indicator corresponding to a change of resistance of the sensing element relative to a reference resistance that defines whether threshold abrasion condition is satisfied. The abradable resistive material has a higher resistivity than a conductive material from which the at least one sensing element is composed.
[0066] In some embodiments, a change of the cross-sectional length of the abradable resistive material relative to a reference length causes a change to the conductivity measurement as a function of contact area between the at least one sensing element and a corresponding return sensor.
[0067] In some embodiments, a return element is positioned on the first contact face of the first body as the sensing element and coupled to the second terminal of the power source, wherein the second body is non-conductive. In some embodiments, an abrasion sensing element is attached to the sensing element and the return element on the first body.
[0068] In some embodiments, a least one sensing element among the plurality of sensing elements is entrenched into a grove of a second face of the first body that is different from the first contacting face of the first body. A portion of the at least one sensing element is exposed outside the first body.
[0069] It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more components, whether or not those components are in physical contact with one another. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.
[0070] The description in the present disclosure should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).
[0071] While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. An apparatus comprising: a power source including a first terminal and a second terminal; at least one of: an ammeter connected to the first terminal of the power source to measure electrical current at the first terminal; or a voltmeter connected in parallel with the first terminal and the second terminal of the power source to measure voltage across the power source; and a plurality of electrically conductive sensing elements is coupled to the first terminal of the power source and positioned on a first contact face of a surface of an external first body, flush with the first contact face, and arranged in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS) that generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the external first body at the spatial location of the NISMS, wherein the measurements from the at least one of the ammeter or the voltmeter are output as a contact event indicator that includes a conductivity measurement corresponding to the electrical signal generated by the physical contact at a sequence of the NISMSs with the conductive area on opposing contact face.

2. The apparatus of claim 1, wherein: the pattern includes an array including a first row, a second row, a first column, and a second column; and the conductivity measurement corresponding to first and second NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a first axis of rotation; the conductivity measurement corresponding to first and third NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a second axis of rotation orthogonal to the first axis of rotation; the conductivity measurement corresponding to third and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a third axis parallel to the first axis of rotation; and the conductivity measurement corresponding to second and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a fourth axis parallel to the third axis.

3. The apparatus of claim 1, further comprising: a set of return elements asymmetrically arranged in the pattern and positioned on the opposing contact face of the second external body, and including a continuity return element characterized by a wider diameter than non-aligned return elements among the set of return elements, wherein: the plurality of sensing elements is asymmetrically arranged in the pattern; wherein when the opposing contact face physically contacts the first contact face, the continuity return element maintains electrical contact with a first continuity sensing element among the plurality of sensing elements while any non-aligned return element electrically contacts a corresponding non-aligned sensing element as a non-aligned pair; the contact event indicator output includes a first indicator of translation in a first direction along a first axis when a first non-aligned pair are contacting; the contact event indicator output includes a second indicator of translation in a second direction along the first axis when a second non-aligned pair are contacting, the first and second directions opposite to each other; the contact event indicator output includes a third indicator of translation in a third direction along a second axis orthogonal to the first axis when a third non-aligned pair are contacting; and the contact event indicator output includes a fourth indicator of translation in a fourth direction along a fourth axis orthogonal to the second axis when a fourth non-aligned pair are contacting, the third and fourth directions opposite to each other.

4. The apparatus of claim 3, wherein contact event indicator output is used to determine an angle of translation and distance of translation from an origin where the spatial locations of the continuity sensing element and the continuity return element overlap, based on: at least one of the first, second, third, or fourth indicators; and the origin relative the spatial locations of the sequence of the NISMSs among the non-aligned pairs contacting.

5. The apparatus of claim 1, further comprising an abrasion sensing element attached to at least one sensing element among the plurality of sensing elements, where the abrasion sensing element is composed of an abradable resistive material that reduces cross-sectional length in response to abrasion from physical contact with the second external body, wherein the contact event indicator output includes an abrasion indicator corresponding to a change of resistance of the sensing element relative to a reference resistance.

6. The apparatus of claim 5, wherein the abradable resistive material has a higher resistivity than a conductive material from which the at least one sensing element is composed.

7. The apparatus of claim 5, wherein a change of the cross-sectional length of the abradable resistive material relative to a reference length causes a change to the conductivity measurement as a function of contact area between the at least one sensing element and a corresponding return sensor.

8. The apparatus of claim 1, further comprising a return element positioned on the first contact face of the external first body as the sensing element and coupled to the second terminal of the power source, wherein the second external body is non-conductive.

9. The apparatus of claim 8, further comprising an abrasion sensing element attached to the sensing element and the return element on the external first body.

10. The apparatus of claim 1, wherein a least one sensing element among the plurality of sensing elements is entrenched into a grove of a second face of the first body that is different from the first contacting face of the first body, wherein a portion of the at least one sensing element is exposed outside the first body.

11. A method comprising: providing a power source including a first terminal and a second terminal; providing at least one of: an ammeter connected to the first terminal of the power source to measure electrical current at the first terminal; or a voltmeter connected in parallel with the first terminal and the second terminal of the power source to measure voltage across the power source; and providing a plurality of electrically conductive sensing elements coupled to the first terminal of the power source and positioned on a first contact face of a surface of an external first body, flush with the first contact face, and are arranged in a pattern such that a spatial location of each sensing element among the plurality of electrically conductive sensing elements forms a normal interface surface mating sensor (NISMS) that generates an electrical signal when a conductive area on an opposing contact face of an external second body physically contacts the first contact face of the first body at the spatial location of the NISMS, outputting the measurements from the provided at least one of the ammeter or the voltmeter as a contact event indicator that includes a conductivity measurement corresponding to the electrical signal generated by the physical at a sequence of the NISMSs with the conductive area on opposing contact face.

12. The method of claim 11, wherein: the pattern includes an array including a first row, a second row, a first column, and a second column; and the conductivity measurement corresponding to first and second NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a first axis of rotation; the conductivity measurement corresponding to first and third NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a second axis of rotation orthogonal to the first axis; the conductivity measurement corresponding to third and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a third axis parallel to the first axis; and the conductivity measurement corresponding to second and fourth NISMSs as the sequence of the NISMSs in the physical contact with the second external body indicates angular misalignment about a fourth axis parallel to the third axis.

13. The method of claim 11, further comprising: providing a set of return elements asymmetrically arranged in the pattern and positioned on the opposing contact face of the second external body, and including a continuity return element characterized by a wider diameter than non-aligned return elements among the set of return elements, wherein: the plurality of sensing elements is asymmetrically arranged in the pattern; wherein when the opposing contact face physically contacts the first contact face, the continuity return element maintains electrical contact with a first continuity sensing element among the plurality of sensing elements while any non-aligned return element electrically contacts a corresponding non-aligned sensing element as a non-aligned pair;

the contact event indicator output includes a first indicator of translation in a first direction along a first axis when a first non-aligned pair are contacting; the contact event indicator output includes a second indicator of translation in a second direction along the first axis when a second non-aligned pair are contacting, the first and second directions opposite to each other; the contact event indicator output includes a third indicator of translation in a third direction along a second axis orthogonal to the first axis when a third non-aligned pair are contacting; and the contact event indicator output includes a fourth indicator of translation in a fourth direction along a fourth axis orthogonal to the second axis when a fourth non-aligned pair are contacting, the third and fourth directions opposite to each other.

14. The method of claim 13, further comprising: determining, based on the contact event indicator output, an angle of translation and distance of translation from an origin where the spatial locations of the continuity sensing element and the continuity return element overlap, based on: at least one of the first, second, third, or fourth indicators; and the origin relative the spatial locations of the sequence of the NISMSs among the non-aligned pairs contacting.

15. The method of claim 11, further comprising: providing an abrasion sensing element attached to at least one sensing element among the plurality of sensing elements, where the abrasion sensing element is composed of an abradable resistive material that reduces cross-sectional length in response to abrasion from physical contact with the second external body, wherein outputting the contact event indicator includes outputting 1 an abrasion indicator corresponding to a change of resistance of the sensing element relative to a reference resistance.

16. The method of claim 15, wherein the abradable resistive material has a higher resistivity than a conductive material from which the at least one sensing element is composed.

17. The method of claim 15, wherein a change of the cross-sectional length of the abradable resistive material relative to a reference length causes a change to the conductivity measurement as a function of contact area between the at least one sensing element and a corresponding return sensor.

18. The method of claim 11, further comprising a return element positioned on the first contact face of the first body as the sensing element and coupled to the second terminal of the power source, wherein the second external body is non-conductive.

19. The method of claim 18, further comprising an abrasion sensing element attached to the sensing element and the return element on the first body.

20. The method of claim 11, wherein a least one sensing element among the plurality of sensing elements is entrenched into a grove of a second face of the first body that is different from the first contacting face of the first body, wherein a portion of the least one sensing element is exposed outside the first body.